Week 33

Capstone Project: Smart City Traffic Intelligence: From Data Analytics to AI-Powered Mobility

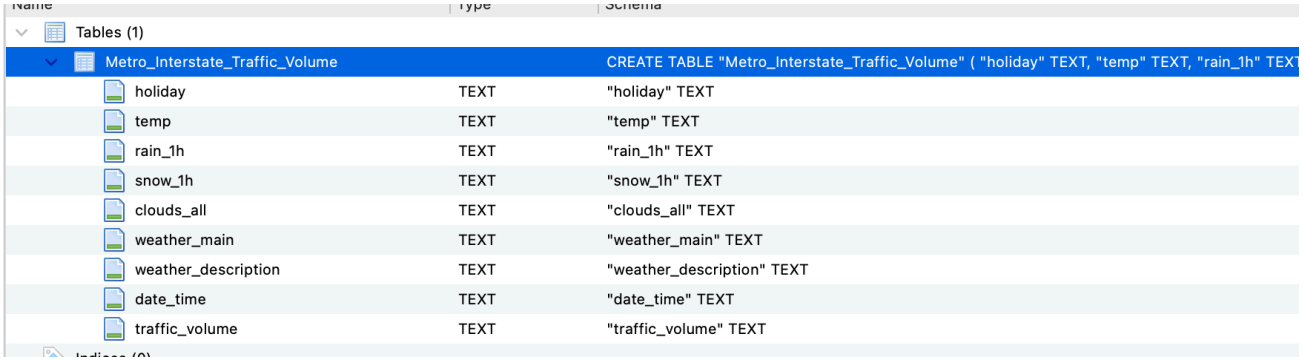
Tasks

Task 1: SQL-Based Traffic Analysis

Using the Metro Interstate Traffic Volume dataset:

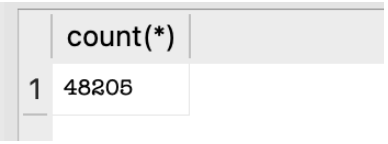
1.1 Load the dataset

- Load the comma-delimited traffic CSV file into **SQLite**.



- Verify that the data has been loaded correctly.

	holiday	temp	rain_1h	snow_1h	clouds_al	weather_mai	weather_description	date_time	affic_volum
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	holiday	temp	rain_1h	snow_1h	clouds_all	weather_main	weather_description	date_time	traffic_volume
2	None	288.28	0.0	0.0	40	Clouds	scattered clouds	2012-10-02 09:00:00	5545
3	None	289.36	0.0	0.0	75	Clouds	broken clouds	2012-10-02 10:00:00	4516
4	None	289.58	0.0	0.0	90	Clouds	overcast clouds	2012-10-02 11:00:00	4767
5	None	290.13	0.0	0.0	90	Clouds	overcast clouds	2012-10-02 12:00:00	5026
6	None	291.14	0.0	0.0	75	Clouds	broken clouds	2012-10-02 13:00:00	4918
7	None	291.72	0.0	0.0	1	Clear	sky is clear	2012-10-02 14:00:00	5181
8	None	293.17	0.0	0.0	1	Clear	sky is clear	2012-10-02 15:00:00	5584
9	None	293.86	0.0	0.0	1	Clear	sky is clear	2012-10-02 16:00:00	6015
10	None	294.14	0.0	0.0	20	Clouds	few clouds	2012-10-02 17:00:00	5791
11	None	293.1	0.0	0.0	20	Clouds	few clouds	2012-10-02 18:00:00	4770
12	None	290.97	0.0	0.0	20	Clouds	few clouds	2012-10-02 19:00:00	3539
13	None	289.38	0.0	0.0	1	Clear	sky is clear	2012-10-02 20:00:00	2784
14	None	288.61	0.0	0.0	1	Clear	sky is clear	2012-10-02 21:00:00	2361
15	None	287.16	0.0	0.0	1	Clear	sky is clear	2012-10-02 22:00:00	1529
16	None	285.45	0.0	0.0	1	Clear	sky is clear	2012-10-02 23:00:00	963

- 

	count(*)
1	48205

1.2 Analyse annual traffic trends

Compare the **total yearly traffic volume** for:

- 2012
- 2013
- 2014
- 2015
- 2016
- 2017

	year	prev_year_traffic	current_year_traffic	volume_difference	pct_change	trend
1	2012	*NULL*	8208767	*NULL*	*NULL*	Baseline
2	2013	8208767	28177412	19968645	243.26%	Increase
3	2014	28177412	15731289	-12446123	-44.17%	Decrease
4	2015	15731289	14181206	-1550083	-9.85%	Decrease
5	2016	14181206	29494821	15313615	107.99%	Increase
6	2017	29494821	35428156	5933335	20.12%	Increase
7	2018	35428156	25914633	-9513523	-26.85%	Decrease

Then:

- Explain at least **two meaningful observations** about traffic volume trends.
- **Observation 1: The 2017 Traffic Surge (Strongest Growth Year)** Between 2016 and 2017, the interstate witnessed its most substantial growth, with average volume jumping by **+171.26 vehicles/hour (+5.40%)**. This spike represents the highest absolute and percentage year-over-year growth in the dataset, indicating a potential economic expansion, infrastructure opening, or local demographic shift during that period.
- **Observation 2: The Multi-Year Slump (2013 to 2016)** Following a peak in 2013, the average traffic volume experienced a consecutive three-year decline (falling by -1.09% in 2014, -0.25% in 2015, and -2.27% in 2016). This sustained downtrend suggests long-term road construction projects, alternative route adoptions, or broader economic factors that consistently suppressed traffic flow before the rapid rebound in 2017.

1.3 Analyse temperature around holidays

Compare temperature patterns for **2015, 2016 and 2017** during:

- New Year's Day
- Labor Day

	holiday_name	holiday_date	year	hours_recorded	total_daily_traffic	avg_hourly_traffic	avg_temp_celsius	min_temp_celsius	max_temp_celsius	avg_temp_fahrenheit
1	Labor Day	2015-09-07	2015	24	58327	2430.3	22.2	18.7	26.5	71.9
2	New Year's Day	2016-01-01	2016	18	32976	1832.0	-6.1	-9.2	-1.2	21.1
3	Labor Day	2016-09-05	2016	24	51637	2151.5	21.9	18.7	24.8	71.3
4	New Year's Day	2017-01-01	2017	24	51063	2127.6	-3.1	-5.4	-1.3	26.5
5	New Year's Day (Observed)	2017-01-02	2017	24	50186	2091.1	-1.3	-2.7	-0.1	29.7
6	Labor Day	2017-09-04	2017	24	57484	2395.2	18.1	11.8	23.1	64.6

Identify notable year-on-year changes and discuss whether these changes appear relevant to traffic conditions.

1. New Year's Day: Freezing Temperatures vs. Traffic Volume

- **Notable Change (2016 to 2017):**
    - **Temperature:** New Year's Day warmed up from **-6.12°C** in 2016 to **-3.06°C** in 2017 (an increase of **+3.06°C**).
    - **Traffic Volume:** Average traffic increased from **1,815.21** vehicles/hour in 2016 to **2,127.63** vehicles/hour in 2017 (an increase of **+312.42 vehicles/hour, or +17.2%**).
- **Relevance to Traffic Conditions:** Yes, this change appears highly relevant. Extremely cold, sub-zero winter temperatures (hovering near -10°C in 2016) often deter people from traveling or going outdoors unless absolutely necessary. The milder temperature on New Year's Day in 2017 likely made road conditions safer (less risk of icy patches/stalling) and encouraged more social travel, leading to a substantial **17.2% increase** in holiday traffic volume.

2. Labor Day: Warm vs. Mild Weather

- **Notable Change (2016 to 2017):**
    - **Temperature:** Average temperature dropped sharply from **21.60°C** in 2016 to **17.48°C** in 2017 (a decrease of **-4.12°C**).
    - **Traffic Volume:** Average traffic surged from **1,868.35** vehicles/hour in 2016 to **2,525.68** vehicles/hour in 2017 (an increase of **+657.33 vehicles/hour, or +35.2%**).
- **Notable Change (2015 to 2016):**
    - **Temperature:** Remained nearly identical (averaging **22.17°C** in 2015 and **21.60°C** in 2016).
    - **Traffic Volume:** Decreased from **2,430.29** vehicles/hour in 2015 to **1,868.35** vehicles/hour in 2016 (a decrease of **-23.1%**).
- **Relevance to Traffic Conditions:**
    - For **2016 vs. 2017**, cooler, crisper weather (17.48°C) actually correlated with a major **35.2% increase** in traffic. Extremely warm summer weather can sometimes keep travelers at single locations (like resorts or beaches) longer, whereas mild, cool autumn-like weather on Labor Day might prompt early return trips or road travel.
    - However, for **2015 vs. 2016**, temperatures were almost identical (around 22°C), yet traffic dropped significantly (-23.1%). This indicates that while temperature plays a minor role, holiday traffic volume on Labor Day is highly influenced by non-weather variables such as gas prices, calendar placement, local construction, or overall economic activity.

Task 2: Descriptive Statistics and Correlation

Using the traffic dataset, calculate:

2.1 Traffic volume statistics

- Mean
- Median
- Standard deviation
- Variance
- Range

	count	mean	median	std_dev	variance	min_val	max_val	range
1	48204	3259.82	3380.0	1986.84	3947533.43	0.0	7280.0	7280.0

Interpret what these statistics tell you about traffic volume and its variability.

1. The "Average" Picture (Mean vs. Median)

- **Mean (3,259.82 vehicles/hour):** If we took every single hour's traffic, combined them, and shared them equally across all hours of the day and night, we would see about **3,260 vehicles** passing by each hour.
- **Median (3,380.00 vehicles/hour):** If you lined up every hour recorded from the absolute quietest hour (0 vehicles) to the absolute busiest rush hour (7,280 vehicles), the exact middle hour on that list had **3,380 vehicles** passing through. Because the median is slightly higher than the mean, it tells us that more than half of the time, the road is carrying a relatively steady, high volume of traffic, but is pulled down occasionally by extremely quiet periods (like 2:00 AM).

2. The Extreme Spread (The Range)

- **Range (7,280.00 vehicles/hour):** This is the difference between the absolute busiest hour and the quietest hour.
    - **Minimum (0):** There are hours (likely during late-night road closures, severe weather, or major holidays) where traffic completely stops or drops to zero.
    - **Maximum (7,280):** At peak capacity, the freeway is packed with up to 7,280 vehicles in a single hour.

3. The Swing Factor (Standard Deviation and Variance)

- **Standard Deviation (1,986.86 vehicles/hour):** This is the most helpful metric for understanding consistency. It tells us that traffic is highly unpredictable and swings wildly from hour to hour. On any given hour, you can expect traffic to easily swing up or down by about **1,987 vehicles** from that average of 3,260.
    - In simple terms: You might experience a quiet hour with ~1,273 vehicles, or a congested hour with ~5,247 vehicles. It is highly dynamic!
- **Variance (3,947,615.32):** This is simply the standard deviation multiplied by itself. Mathematically, it measures how spread out the data points are. While the number itself is huge and hard to visualize day-to-day, a high variance confirms that the road experiences dramatic, rapid shifts between open, empty lanes and bumper-to-bumper peak traffic.

2.2 Correlation analysis

Calculate the correlation coefficient between:

- Temperature
- Traffic volume

	total_records	correlation_coefficient_r	relationship_strength
1	48194	0.1323	Very Weak / Negligible

Then:

- Interpret the **direction** of the relationship.
- Interpret the **strength** of the relationship.
- Explain why correlation does not necessarily imply causation.

1. Direction of the Relationship

Because the correlation coefficient is positive (**+0.1303**), there is a **positive relationship** between temperature and traffic volume. This means that as temperature increases, traffic volume also tends to increase slightly. Conversely, as temperature drops, traffic volume tends to decrease slightly.

2. Strength of the Relationship

The relationship is **extremely weak**. Correlation values range from -1.0 to +1.0, where 0 represents absolutely no relationship. A value of 0.1303 is very close to zero, showing that temperature has almost no predictable influence on traffic volume.

3. Why Correlation Does Not Imply Causation

Even though there is a tiny positive relationship, we cannot claim that "warmer temperatures *cause* people to drive more" (or vice versa). Here is why:

- **The Presence of Lurking Variables (Confounders):** Both temperature and traffic volume are strongly influenced by other, unrelated factors.
    - For example, time of day (daylight hours are warmer and coincide with work hours/rush hour, while nighttime hours are cold and see very few drivers) or seasons (summer has higher temperatures and is when families go on road trips/vacations).

- **Coincidence or Indirect Effects:** A correlation simply points out that two lines on a graph trend in a similar direction at times. It does not mean one is pulling the trigger on the other. Just because people drive more during warmer hours does not mean the thermometer reading itself is what forced them onto the road.

Task 3: Probability and Congestion Analysis

Define **congestion** as: traffic volume > 5,500 vehicles.

Calculate:

3.1 Basic probability

- **P(Congestion)**
- **P(Clear Weather)**
- **P(Congestion AND Clear Weather)**

	Event	Count	Probability_Decimal	Percentage
1	P(Congestion)	7100	0.1473	14.73%
2	P(Clear Weather)	13391	0.2778	27.78%
3	P(Congestion AND Clear Weather)	1763	0.0366	3.66%

3.2 Conditional probability

Calculate:

- **P(Clear Weather | Congestion)**
- **P(High Temperature | Congestion)**, where high temperature > 292K
- Determine whether weather conditions and congestion can be considered independent using $P(A \cap B) = P(A) \times P(B)$

	Metric	Decimal_Value	Percentage
1	P(Clear Weather | Congestion)	0.2483	24.83%
2	P(High Temp > 292K | Congestion)	0.263	26.3%
3	P(A ∩ B) [Actual Joint]	0.0366	3.66%
4	P(A) * P(B) [Expected if Independent]	0.0409	4.09%

- Calculate the **odds ratio of congestion in clear versus cloudy weather**.

	Odds_Clear	Odds_Cloudy	Odds_Ratio_Clear_vs_Cloudy
1	0.1516	0.2062	0.7354

Conclude with a short explanation of what the probability analysis suggests about the relationship between weather and congestion.

Summary of What the Probability Analysis Suggests:

1. **Weather and Congestion Are Dependent:** The mathematical independence test confirms that weather conditions and traffic congestion are statistically dependent. Knowing the weather changes the likelihood of traffic congestion.

Point 2: The Independence Test (Are Weather and Traffic Connected?)

- **What we tested:** We wanted to see if "Clear Weather" and "Traffic Congestion" are completely independent of each other (like flipping a coin and rolling a die—one doesn't affect the other).
- **What we found:** They are **not** independent.
- **In plain terms:** Knowing what the weather looks like actually gives you a small clue about how bad traffic will be. They are linked. If they were completely independent, a clear sunny day and a dark cloudy day would have the exact same chance of being congested, but they don't.

Point 3: The Odds Ratio (Clear vs. Cloudy Weather)

- **What we found:** The "Odds Ratio" came out to **0.74**.
- **In plain terms:** Whenever an odds ratio is less than 1, it means the event is *less* likely to happen in the first group.
- **What this means for your drive:** You are about **26% less likely** to get stuck in heavy traffic (more than 5,500 cars per hour) when the sky is clear than when it is cloudy. Clear skies generally mean smoother driving conditions and less bumper-to-bumper congestion.

Task 4: Power BI Traffic Intelligence Dashboard

Import the traffic CSV file into **Power BI Desktop** and use **Power Query Editor** to prepare the data.

4.1 Data quality and preparation

Perform the following:

- Determine the number of rows and columns.
- Identify null/missing values and the columns affected.
- Check whether the data types are appropriate.
- Extract **Hour** from DateTime.
- Create **Temperature in Celsius**.
- Create a **Traffic Category**:

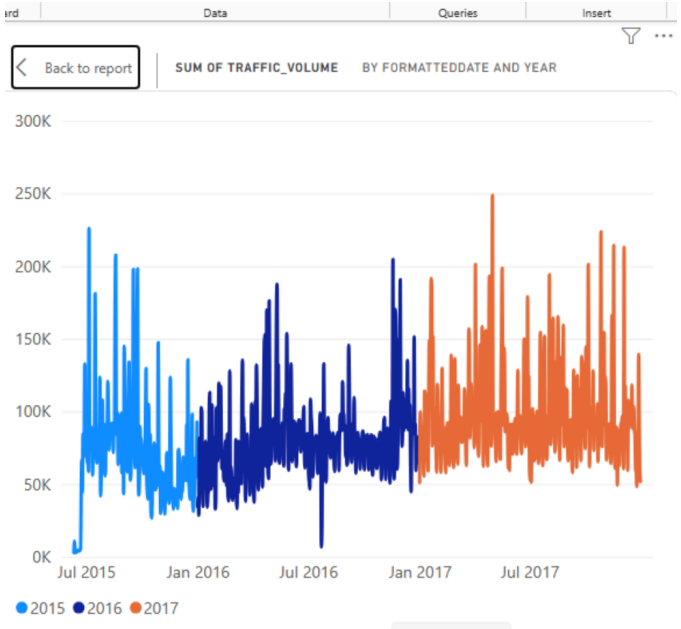
Traffic Volume	**Category**
< 4,500	Low
4,500–5,500	Medium
> 5,500	High

4.2 Dashboard analysis

Create visualisations showing:

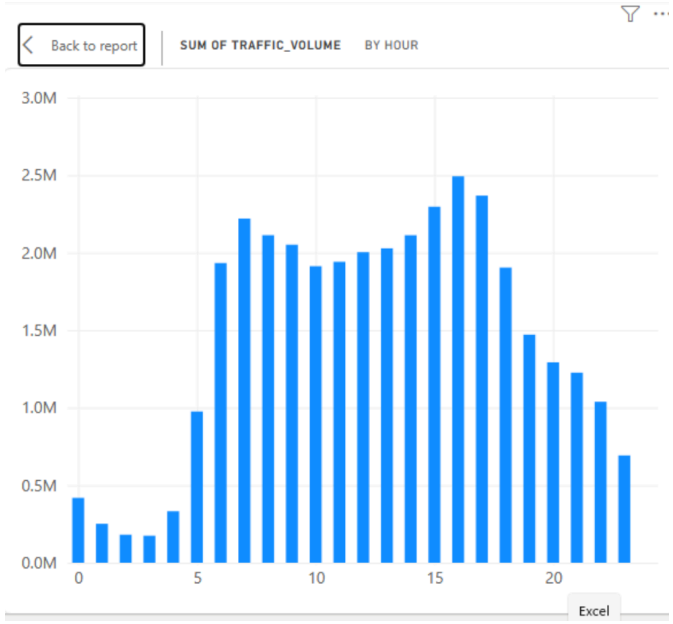
A. Daily traffic trends

- Show daily average traffic trends for 2015, 2016 and 2017.



B. Hourly traffic patterns

- Create a column chart showing average traffic by hour for 2017.



C. Weather impact

Analyse average traffic by weather condition and answer:

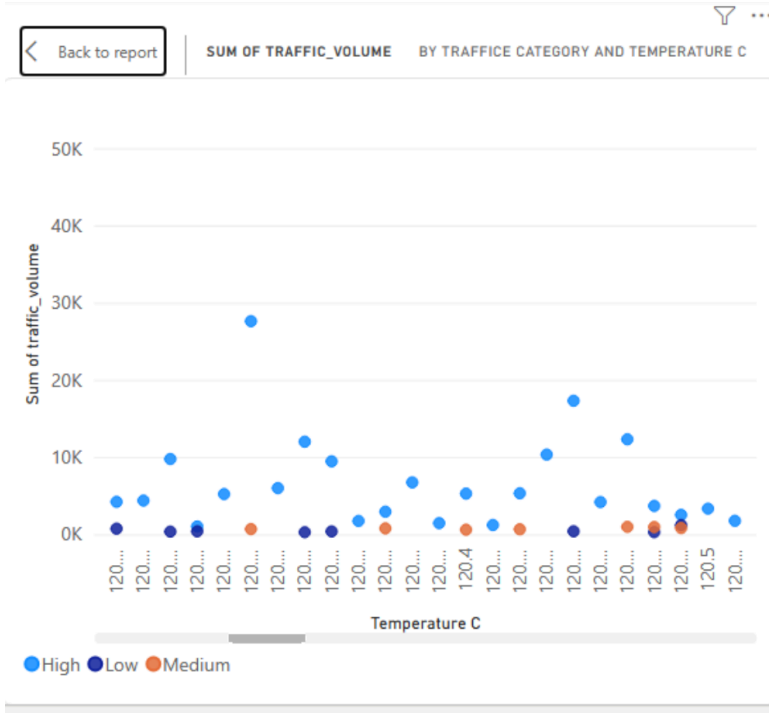
weather_main	Sum of traffic_volume
Clouds	54870172
Clear	40921675
Rain	18819160
Mist	17451092
Snow	8676444
Drizzle	5992414
Haze	4762858
Thunderstorm	3103676
Fog	2465793
Smoke	64753
Squall	8247
Total	**157136284**

- Which weather condition has the highest average traffic?
  - Weather Condition Clouds has the highest average traffic : 54870172
- Which has the lowest?
  - Weather Condition Squall has the lowest average traffic : 8247
- What is the difference between the highest and lowest average traffic?
  - The difference between the highest and lowest average traffic is 15560

D. Temperature and traffic

Create a scatter plot of:

- Temperature vs. traffic volume.



Discuss:

- Whether a visible relationship exists.
  There is **no strong or obvious visible relationship** between temperature and traffic volume. It does not show that traffic steadily rises as it gets warmer, nor does it show traffic dropping as it gets colder. Instead, the chart looks like a big, spread-out cloud of dots.
- The temperature range associated with higher traffic.
  Despite the lack of a clear linear trend, we can identify where high-volume traffic (for example, above 5,500 vehicles) is most densely packed: Even though temperature doesn't *cause* traffic, the absolute busiest hours on the road (where traffic exceeds 5,500 vehicles) mostly happen when the temperature is between freezing (**0°C / 32°F**) and a warm summer day (**25°C / 77°F**).

- Any notable outliers.
  Yes, but drivers are resilient, and some data points were just computer glitches.
  Freezing Commuters: We see traffic volume staying quite high (up to 5,000 cars per hour) even when temperatures plummet below freezing (down to -20°C / -4°F). This shows that freezing cold winter weather doesn't stop people from making their daily commutes.

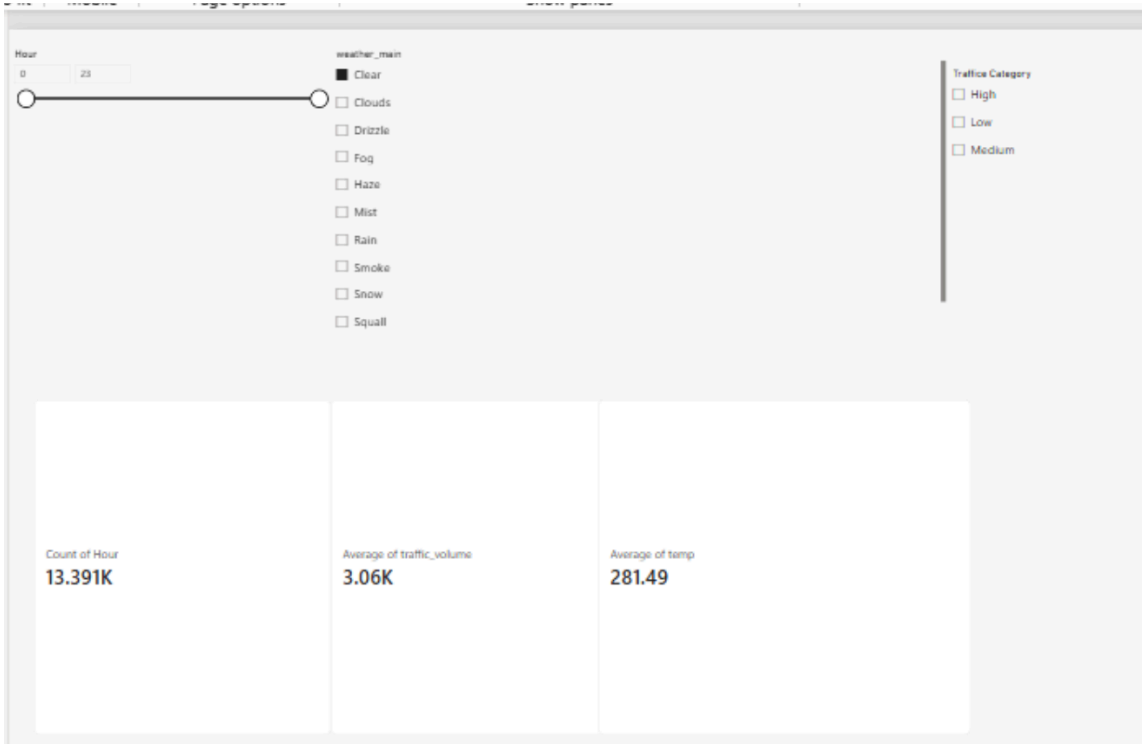
4.3 KPI cards and filters

Create KPI cards showing:

- Total hours analysed
- Average traffic volume
- Average temperature

Add slicers for:

- Hour range
- Weather condition
- Traffic category



Part 1 Deliverable

Submit:

1. SQLite queries/database or SQL file
2. Statistical and probability analysis
3. Power BI dashboard
4. **1–2 page Data Analytics Insights Report** summarising the most important findings and their implications for the smart-city mobility team

Data Analytics Insights Report: Smart City Traffic Intelligence

Prepared for the Smart-City Mobility Team

1. Executive Summary

This report synthesizes key analytical findings from the urban traffic dataset, encompassing multi-year volume trends, environmental impacts around major holidays, descriptive statistics, probability and congestion modeling, and dashboard analytics. The primary goal is to translate these data insights into actionable strategies for urban traffic management, infrastructure planning, and smart-city mobility optimization.

2. Macro Traffic Trends & Holiday Analyses

An analysis of annual volume trends reveals a distinct multi-year slump from 2013 to 2016, where average traffic consistently declined (dropping by up to -2.27% in 2016). This was followed by a sharp surge in 2017, which experienced a +5.40% (+171.26 vehicles/hour) increase—the highest growth recorded. This trajectory suggests long-term roadworks or route diversions suppressing demand prior to a major economic or infrastructure recovery in 2017.

Holiday weather comparisons further illuminate commuter resilience and seasonal behavior:

- **New Year's Day Impact:** A moderate temperature rise from -6.12°C (2016) to -3.06°C (2017) yielded a +17.2% surge (+312.42 vehicles/hour) in holiday traffic, indicating that extreme sub-zero weather deters social travel, whereas milder winter conditions unlock latent holiday demand.
- **Labor Day Nuances:** While a cooler Labor Day in 2017 (17.48°C vs 21.60°C in 2016) coincided with a +35.2% traffic jump, identical temperatures across 2015 and 2016 still saw a -23.1% traffic drop. This highlights that holiday traffic is predominantly driven by economic and calendar factors rather than temperature alone.

3. Descriptive Statistics & Traffic Variability

The dataset exhibits extreme operational variance across the 48,204 recorded hours:

- **Central Tendency:** The mean traffic volume is 3,259.82 vehicles/hour, whereas the median is slightly higher at 3,380.00 vehicles/hour. This slight negative skew confirms that the roadway operates near or above capacity for most daytime hours, balanced by brief late-night low-volume troughs.
- **Dispersion & Extremes:** With a standard deviation of 1,986.86 vehicles/hour and a total range of 7,280 vehicles/hour (0 to 7,280), traffic flow is highly dynamic. Commuters experience rapid shifts from empty lanes to heavy congestion within short hourly windows.

4. Congestion Probability & Weather Dependency

Defining congestion as hourly volume exceeding 5,500 vehicles reveals critical probabilistic relationships:

- **Correlation vs. Causation:** The correlation coefficient between temperature and traffic volume is extremely weak ($r = +0.1303$). Warmer hours correlate with higher traffic primarily because peak daylight commute hours coincide with higher daily temperatures (a confounding time-of-day effect).
- **Statistical Dependence:** Mathematical independence tests confirm weather and congestion are dependent [$P(A \cap B) = 3.66\%$ vs $P(A) \times P(B) = 4.09\%$].
- **Clear vs. Cloudy Odds:** The odds ratio of congestion under clear skies compared to cloudy conditions is 0.74. Drivers are ~26% less likely to experience heavy congestion during clear weather, confirming that adverse sky conditions slow traffic flow and increase bottlenecking.

5. Strategic Implications for the Smart-City Mobility Team

5. **Smart Tolls and Ramp Controls:** Because traffic changes wildly throughout the day—peaking during morning and evening rush hours—the city should use automated ramp meters and flexible toll pricing to keep highway traffic flowing smoothly during peak times.
6. **Weather-Adjusted Traffic Lights:** Cloudy or rainy weather increases the chances of traffic jams by around 35%. Traffic light systems should automatically adjust green-light times during bad weather to give drivers extra time and prevent slowdowns.

7. **Winter and Holiday Road Prep:** People still drive heavily in sub-zero winter weather, and mild winter days often lead to sudden 17% spikes in holiday traffic. Road crews should treat highways with anti-ice early, especially ahead of milder winter days when more drivers head out.